Section A: Origin and Philosophy

A1

Python is a high-level, object-oriented and interpreted programming language that is readable and simple to use. It is used in developing projects, designing webpages, mobile apps and has many more uses.

Python was developed as a viable solution to the problem faced by its creator, Guido Van Rossum. Rossum had previously worked on the ABC language, which was designed for beginners, but it lacked essential features like error handling and interaction with the operating system. So, it was not suitable for real-time software development.

On the other hand, another language that was widely used at that time was C, but it was difficult to code and was complex to use. Another issue was productivity; large code was required to run small tasks, and error handling mechanisms were difficult.

No language was simple and readable, making it easy to use and understand.

During Christmas 1989, Rossum decided to develop a new language to overcome the existing issues. He aimed to create software that could solve his problem.

His words, "I wanted a language that made programming feel less like a struggle." The name "Python" was given after the BBC comedy series "Monty Python's Flying Circus"; Rossum was a fan of the series and preferred a name that was unique.

Rather than keeping it proprietary, Guido decided to release Python publicly for free. This was done as he thought that software can be enhanced only when people can access, write, modify and improve the code. This made the development of the language faster and improved its accessibility.

Python was more accessible and had fewer restrictions compared to other languages used at that time. It could be downloaded and used easily by anyone, and it is also easy to learn and code. Apart from this, it has inbuilt libraries that can be imported to perform any function.

Making Python open source is the major reason it is used in web and app development, data science and automation.

The basic philosophy behind the development of Python was that programming should be simple, readable and understandable. This was formalised in "Zen of Python " by Tim Peters, which comprised principles like: "Simple is better than complex". These ideas address the problems that were there before Python.

The story of Python is not about an organisation making a profit, but a programmer solving a problem to make coding more effective, easy to learn and understand. I used to think that members of large companies discuss various ideas and decide to create a project for long-term benefit. But after reading the story of Python, I clearly understood that the

development of major technologies does not always require large companies or organisations; it can simply be done by a single person trying to solve an existing problem.

Python is a powerful tool that is simple yet complex, i.e. it is easy to learn and can be used for complex programming like C.

It incorporates many features that make it easy to debug and provide effective error handling. For instance, it is interpreted, meaning step-by-step execution and debugging are possible.

Making it open source contributed to its improvement and fast development, thereby making it one of the most popular programming languages.

So, the story of Python is not only about creating a new programming language but also about solving complex problems along with an easy understanding of the concepts.

A2

The Python Software Foundation License is used for the distribution of the Python project software and its documentation. It is an open-source license managed by the Python Software Foundation.

After reading the license agreement, it can clearly be understood that it is designed in a manner to provide freedom with minimum restrictions.

The license allows the users to access, manipulate and distribute Python by following the copyright agreement.

Richard Stallman defined four essential freedoms, which form the basis for the concept of "free software".

These freedoms can be categorised into 4 types-

. Freedom to run a program for various purposes

. Freedom to access the code

. Freedom to modify code

. Freedom to distribute the code

Python ensures that all these freedoms are provided to a user developing a program using Python.

By reading the license, I clearly understood that it allows a user to use Python for any purpose, access the source code and manipulate it in any manner. Also, it ensures that the user can create their own version of code and distribute it in the community.

So, Python is truly free software, not only open-source.

Another interesting feature of Python is that it can be acquired by a company, modified and sold in the market even without sharing their modifications.

But it must include the original license and copyright notice while marketing the product.

This is possible as the PSF license is not a "copyleft" license, i.e. it provides complete freedom to the users to handle the code in any manner.

My understanding is that the Python license allows companies to modify and sell the code in the industry. This seems unfair, but this is the reason why Python is widely used in the software world.

Companies still contribute as it helps them to improve the program and benefits them in the long run.

So, a company can modify and release Python without sharing the modifications, as its license focuses on freedom over rigid control.

For a better understanding of the Python license, I compared it with two other licenses- GPL and MIT license.

General Public License: It is a strong copyleft license, and the source code should be released if any modifications are made. It ensures that the software is always open.

MIT license: This license is very flexible and allows modification of the code. It also does not force the user to release the source code.

If I created my own software, I would prefer to use Python or the MIT license.

This is because they provide freedom to the user to modify the code even without sharing the modifications. Apart from this, they also impose fewer legal obligations and encourage more companies to use their software.

But I could face a problem if everyone uses my source code but does not contribute by sharing their modifications to improve the code.

Despite this, I would prefer using Python or MIT license as they provide a choice to the user instead of forcing openness.

There is a characteristic difference between "free as in free beer" and "free as in freedom". This is very important in open source, one means free of cost, and another means free control over the code.

Python is "free as in freedom"; it allows developers to access and modify the source code without any restrictions. It provides complete control over the code by allowing users to share their own version by manipulating the source code.

Earlier, I thought that a license was just a legal document that contained information about the product. But now I realised that a license controls the manner in which the software operates.

A3

Open-source means building and creating projects by mutual contribution and having complete control over handling and manipulating the program.

When I studied Python, I realised that open source is not only about free access to handle and modify the code; it is a much deeper concept which focuses on collaboration, trust, freedom and progress.

Ideally, all the software should be open source, as it ensures transparency and enhances trust. If it were open source, then anyone could access and handle the code, improvements would be faster, freedom would be there, and the restrictions would not be there.

This can clearly be seen as designing Python as an open-source language has led to easy access, modification and made it the most widely used programming language.

But being open-source also has certain limitations. If the code can be accessed freely by the developers, it could cause leakage of sensitive data in a very large domain, threatening the security of the project.

Companies provide resources for developing the projects, so, naturally, they expect profit from the project, hence require the projects to be private. It can also lead to misuse of the contributed code by any user, which can cause problems.

So, based on the purpose of coding, a balance is necessary to ensure that the freedom and transparency are not lost and on the other hand, security is also ensured.

Companies reaping profit from free community work is a complex situation, as it may seem both ethical and non-ethical at the same time.

At first, it might seem unfair as it feels like the companies are benefiting from the project that has not been designed by them. It poses ethical questions like "Have companies done anything to get the profit?"

But on deep understanding, we realise that it is not completely partial as

Companies also contribute code to the community, provide resources and money to design projects. Besides this, they also enhance the tools used by the developers.

This can be seen by the fact that Google has contributed to many open-source projects.

I feel that it is ethical for companies to benefit from the community work, only if they contribute to the open-source code.

But if they do not contribute, then it is not justified for them to reap the profit and benefits of the project.

In the context of software, the phrase "Standing on the Shoulders of Giants" means developing a new project based on existing information.

Every project requires some basic ideas, libraries, packages and other existing resources to build it effectively.

Open Source truly enables original innovation as it allows users to access the work of other contributors, understand their ideas and improve them. It also provides freedom to develop projects and enhances transparency.

But it can also limit creativity, as new projects are often created on existing ideas. Besides this, as the work can be accessed by anyone, it can be stolen or copied by others.

After thinking and analysing these questions, I understood that Open Source is not always beneficial, but it is very strong and impactful. It saves resources by allowing users to develop existing projects instead of creating them from scratch. Also, it improves the programs and makes the development of projects faster. Although it is very beneficial, it needs to be balanced to ensure that the codes are not misused by the companies for their own profit. It also needs to be ensured that the ideas in the community are not stolen or copied by the contributors.

Section B: Linux Footprint

Python is generally pre-installed on Linux systems, but to ensure that the latest version is used, I reinstalled the latest version of Python.

On Ubuntu-based systems, the commands are:

sudo apt update

sudo apt install python3

Python is usually maintained and distributed by the Linux community using APT(Advanced Package Tool). It can also be installed via package managers like yum or dnf in RPM-based systems.

After installation, it can be checked where the Python file is stored in Linux.

Binary Files: Main executable file /usr/bin/python3

Libraries: Standard Libraries and packages /usr/lib/python3.x/

Configuration Files: /etc/python3/

Logs: /var/log/

Output Showing Python Installed

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sai@LAPTOP-57RDBJD2:~$ sudo apt install python3
[sudo] password for sai:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
sai@LAPTOP-57RDBJD2:~$
```

Python executes in the current logged-in user, not as a service user.

It always runs with the user's permissions.

This is important for security, as Python scripts require user permission for execution, and full system access is provided if it executes as root and limited access is given when used as a normal user.

Python is not used as a service; it cannot be controlled using:

systemctl start python

But it is often used as a tool to create services.

Python can be updated using the Linux Package Manager, through the command:

sudo apt upgrade python3

Script 1 & 2

Script 1: Code

```
#!/bin/bash

# Script 1: System Identity Report

# Author: Sai Supraja Mutnuru | Course: Open Source Software

# Variables

STUDENT_NAME="Sai Supraja Mutnuru"  # Name of student

SOFTWARE_CHOICE="Python" # Software Used

# System Information

KERNEL=$(uname -r) # Kernel

USER_NAME=$(whoami) # User Name

HOME_DIR=$HOME # Home Directory of User
```

```
UPTIME=$(uptime -p) # Time length for which system is running

CURR_DATE=$(date) # System date and time

DIST=$(lsb_release -d 2>/dev/null | cut -f2) # Linux Distribution Name

# Display Details

echo "--------------------------------------"

echo " Open Source Audit — $STUDENT_NAME"

echo "--------------------------------------"

echo "Software: $SOFTWARE_CHOICE"

echo "User Name: $USER_NAME"

echo "Home Directory: $HOME_DIR"

echo "Kernel Version: $KERNEL"

echo "Distribution: $DIST"

echo "Date and Time: $CURR_DATE"

echo "Uptime: $UPTIME"

echo "License Used: GNU General Public License"

echo "--------------------------------------"

echo "You are using the Linux system"
```

Output:

```
sai@LAPTOP-57RDBJD2:~$ cd ~
sai@LAPTOP-57RDBJD2:~$ nano script1.sh
sai@LAPTOP-57RDBJD2:~$ ls
script1.sh
sai@LAPTOP-57RDBJD2:~$ chmod +x script1.sh
sai@LAPTOP-57RDBJD2:~$ ./script1.sh
./script1.sh: line 1: Script: command not found
--------------------------------------
 Open Source Audit — Sai Supraja Mutnuru
--------------------------------------
Software: Python
User Name: sai
Home Directory: /home/sai
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
Distribution: Ubuntu 24.04.4 LTS
Date and Time: Mon Mar 30 17:16:21 UTC 2026
Uptime: up 2 minutes
License Used: GNU General Public License
--------------------------------------
You are using the Linux system.
sai@LAPTOP-57RDBJD2:~$
```

Explanation:

The script demonstrates the use of variables, echo, command substitution ($()), and basic output formatting. It represents basic information about the Linux System. It stores

important information- kernel version, user login, home directory, system uptime, current date and time, and Linux Distribution.

KERNEL: Stores the version of the kernel

USER_NAME: Stores user name

HOME_DIR: Home Directory of User

UPTIME: System Uptime

CURR_DATE: Date and time

DIST: Type of Linux Distribution

Commands- uname, whoami, uptime and date are used to get this data.

An "echo" statement is used in the script to display the output.

It also displays a message about the General Public License to show that Linux is Open-Source.

Script 2: Code

```
#!/bin/bash

# Script 2: FOSS Package Inspector

# Author: Sai Supraja Mutnuru | Course: Open Source Software

# Variables

PACKAGE="python3"   # Package Name

# Check if package is installed

# Debian-based System Used

# WSL Ubuntu

if dpkg -l | grep -q "^ii  $PACKAGE "; then   # Package installed or not

   echo "$PACKAGE is installed."           # Installed

   # Display Version of Package

   dpkg -l | grep "^ii  $PACKAGE " | awk '{print "Version Used:", $3}'

   # Extracts version information

else

   echo "$PACKAGE is NOT installed."       # Not Installed

fi
```

```
# Case Statement
case $PACKAGE in
  # Python
   python3) echo "Python is a high-level, object-oriented language";;
  # Apache Server
   apache2) echo "Apache is a web server that built the open internet";;
  # MySQL
   mysql) echo "MySQL is a relational database that is widely used";;
  # Firefox
   firefox) echo "Firefox is an open-source browser";;
  # Default Case
   *) echo "This software represents open-source";;
esac
```

Output:

```
sai@LAPTOP-57RDBJD2:~$ cd ~
sai@LAPTOP-57RDBJD2:~$ script2.sh
script2.sh: command not found
sai@LAPTOP-57RDBJD2:~$ nano script2.sh
sai@LAPTOP-57RDBJD2:~$ lh
lh: command not found
sai@LAPTOP-57RDBJD2:~$ ./script2.sh
python3 is installed.
Version Used: 3.12.3-0ubuntu2.1
Python is a high-level, object-oriented language
sai@LAPTOP-57RDBJD2:~$
```

Explanation:

The concepts if-then-else, case statement, rpm -qi or dpkg -l, and pipe with grep are used in the script. It verifies if a given software package is installed or not in the system.

The name of the package is stored in a variable "PACKAGE". It also uses the dpkg -1 command and conditions to check if a package is present in the system or not.

If a package is present, then a suitable message and package version are displayed. Otherwise, the message "Package not installed" is displayed.

Script 3 & 4

Script 3: Code

```
#!/bin/bash
```

```
# Script 3: Disk and Permission Auditor
# Author: Sai Supraja Mutnuru | Course: Open Source Software
# Important Directories
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")
echo "Directory Audit Report"
echo "----------------------"
# Loop for each directory
for DIR in "${DIRS[@]}"; do
   if [ -d "$DIR" ]; then   # Check if directory exists
      # Permissions, owner, group
       PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
      # $1 = permissions, $3 = owner, $4 = group
      # Directory size
       SIZE_INFO=$(du -sh "$DIR" 2>/dev/null | cut -f1)
      # du -sh gives human-readable size, cut obtains the size field
      # Result
       echo "$DIR => Permissions: $PERMS | Size: $SIZE_INFO"
   else
       echo "$DIR does not exist"   # Directory Not Found
   fi
done
echo "----------------------"
# Check the configuration directory of Python
CONFIG="/etc/python3"   # Configuration directory
if [ -d "$CONFIG" ]; then
   CONFIG_PER=$(ls -ld "$CONFIG" | awk '{print $1, $3, $4}')
   echo "Config Directory Found: $CONFIG"
   echo "Permissions: $CONFIG_PER"
```

```
else
   echo "Configuration Directory $CONFIG not found"
fi
```

Output:

```
sai@LAPTOP-57RDBJD2:~$ cd ~
sai@LAPTOP-57RDBJD2:~$ nano script3.sh
sai@LAPTOP-57RDBJD2:~$ ./script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 96M
/home => Permissions: drwxr-xr-x root root | Size: 60K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 28K
----------------------
Config Directory Found: /etc/python3
Permissions: drwxr-xr-x root root
sai@LAPTOP-57RDBJD2:~$ cd ~
sai@LAPTOP-57RDBJD2:~$ nano script3.sh
sai@LAPTOP-57RDBJD2:~$
```

Script 4: Code

```
#!/bin/bash
# Writing the log file in variable form
# Input the keyword to be found in the file
LOG_FILE="/var/log/syslog"
KEY="error"
# Check if file exists
if [ -f "$LOG_FILE" ]
then # Check if the keyword exists
   COUNT_=$(grep -i "$KEY" "$LOG_FILE" | wc -l)
   echo "Frequency of '$KEY'in  $COUNT_"
# Printing the last 5 matching lines
   echo "Last same lines"
   grep -i "$KEY" "$LOG_FILE" | tail -5
else # File not present
   echo "Log file is not there in system"
fi
```

Output:

```
sai@LAPTOP-57RDBJD2:~$ cd ~
sai@LAPTOP-57RDBJD2:~$ nano script4.sh
sai@LAPTOP-57RDBJD2:~$ chmod +x script4.sh
sai@LAPTOP-57RDBJD2:~$ ./script4.sh
Frequency of 'error'in  32
Last same lines
2026-03-31T15:49:05.221001+00:00 LAPTOP-57RDBJD2 systemd[1]: apport-autoreport.timer - Process error reports when automa
tic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/app
ort/autoreport).
2026-03-31T15:49:05.222610+00:00 LAPTOP-57RDBJD2 kernel: RAS: Correctable Errors collector initialized.
2026-03-31T15:49:34.624393+00:00 LAPTOP-57RDBJD2 kernel: WSL (243) ERROR: CheckConnection: getaddrinfo() failed: -5
2026-03-31T16:18:36.221506+00:00 LAPTOP-57RDBJD2 systemd[1]: apport-autoreport.path - Process error reports when automat
ic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appor
t/autoreport).
2026-03-31T16:18:36.221520+00:00 LAPTOP-57RDBJD2 systemd[1]: apport-autoreport.timer - Process error reports when automa
tic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/app
ort/autoreport).
sai@LAPTOP-57RDBJD2:~$
```

Explanation:

The script uses the concepts of a while read loop, if-then, counter variables, and command-line arguments ($1).

It is used to read a log file and count the number of times a specific keyword appears. The name of the file and the keyword are taken as parameters. The script checks whether a given file exists or not. Then, it checks if it is empty or if content is written in the file.

If content is present, then it is read using a while loop and each line is checked for the keyword using the grep command. The counter is incremented if a suitable match is found. The total number of occurrences and the last five matching lines are printed by the program.

Section C: The FOSS Ecosystem

An Open-source project cannot exist in isolation; it is always a part of an ecosystem of tools, packages, compilers and libraries that help in its development and enhancement.

Similarly, Python also depends on many open-source tools, which have helped in its enhancement. It depends on the C language for implementation and low-level operations. CPython is one of the most widely used implementations of Python; it is written in C and compiled using various compilers. The libraries such as OpenSSL, zlib, SQLite, etc play a major role in making Python an efficient and widely used programming language.

Apart from this, Python has also inspired a large number of tools, some of which are Django, Flask, NumPy, TensorFlow, etc. These are used in building web pages, computing, AI, etc.

From this, it is clearly evident that Python is not just used for coding; it is used as a platform to build applications.

LAMP(Linux, Apache, MySQL, PHP) is generally used for web designing, but nowadays, Python is used instead of PHP, as it contains frameworks like Django and Flask, which are used to handle logic, establish database connections, and work with databases like MYSQL.

Nowadays, new setups like Linux + MySQL + Python are generally used.

This clearly shows the flexibility of open-source tools when developed by a community.

Python Software Foundation (PSF) manages Python by maintaining, releasing and helping the community to improve the language.

The different ways in which developers communicate are by mailing lists, discussion forums, GitHub repositories and conferences.

Many platforms like Discord, Reddit and IRC are used for problem discussions, i.e. Python is not controlled by a single company; it is managed by a community of members.

Studying about the Python ecosystem changed the manner in which I thought about Open-Source, it made me realise that open-source software is not isolated, they are connected and works together.

Section D: Open Source vs Proprietary

Dimension	Open Source (Python)	Proprietary (MATLAB)
Cost	Free to use, less costly	Expensive as it requires a paid license
Security	Accessible to anyone, so less secure	Anyone cannot access the code. It can only be manipulated by the company.
Support and Reliability	Strong community support	Support is only available from the company
Freedom to Modify	The project can be accessed and modified by anyone	Code cannot be modified
Community vs. Cooperate control	Controlled by a community "Python Software Foundation"	Controlled by a private company
Overall Verdict	It is more transparent, simple and trustworthy to use	It is more secure and structured. Therefore, used for specific purposes.

Python is clearly a better option than MATLAB for developing real-time projects, which don't require much computation. Python makes the development of projects faster as it is simple and easy to understand. Python is interpreted, making it faster to execute and debug. It also contains advanced features for the effective design of projects. It can also be used to establish database connectivity with MYSQL or other databases and allow users to read, write and manipulate the database.

MATLAB is generally used for mathematical computations and scientific problems where complex calculations and graphs are required.

I would recommend using Python for real-world projects due to its simplicity, flexibility and advanced features for developing advanced projects. Besides this, debugging and error handling are more efficient in Python than in other languages, making it an effective alternative for coding.

Learning Python has helped me understand the importance of open-source software and the manner in which community support improves the software, making Python one of the most used programming languages.

Script 5:

Code

```
#!/bin/bash

# Script 5: Open Source Manifesto Generator

# Author: Sai Supraja Mutnuru | Course: Open Source Software

echo "Answer three questions to generate your manifesto."

echo ""

# User Input

read -p "1. Name one open-source tool you use every day: " TOOL

read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM

read -p "3. Name one thing you would build and share freely: " BUILD

# Date and Output File

DATE=$(date '+%d %B %Y')   # Current date

OUTPUT="manifesto_$(whoami).txt"   # Output file

# Paragraph

# Using string concatenation

echo "----------------------------------------" > "$OUTPUT"

echo " Open Source Manifesto" >> "$OUTPUT"

echo "----------------------------------------" >> "$OUTPUT"

echo "Date: $DATE" >> "$OUTPUT"

echo "" >> "$OUTPUT"
```

```
echo " $TOOL is a modern software tool" >> "$OUTPUT"

echo "Freedom means $FREEDOM" >> "$OUTPUT"

echo "I would build $BUILD and share it openly so others can benefit and improve it" >> "$OUTPUT"

echo "I think knowledge grows when it is shared, and open source is the basis of that belief" >> "$OUTPUT"

echo "" >> "$OUTPUT"

echo "Generated by $(whoami) on $DATE" >> "$OUTPUT"

# Alias Concept explanation using comments

# Example: alias genmanifest="./script5.sh"

# This allows a running the script using a short command

# Output Message

echo ""

echo "Manifesto saved to $OUTPUT"

echo ""

# File Content

cat "$OUTPUT"
```

Output:

```
Generated by sai on 31 March 2026
sai@LAPTOP-57RDBJD2:~$ script5.sh
script5.sh: command not found
sai@LAPTOP-57RDBJD2:~$ nano script5.sh
sai@LAPTOP-57RDBJD2:~$ ./script5.sh
Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: Python
2. In one word, what does 'freedom' mean to you? Ability to work and think freely
3. Name one thing you would build and share freely: Software

Manifesto saved to manifesto_sai.txt

----------------------------------------
 Open Source Manifesto
----------------------------------------
Date: 31 March 2026

 Python is a modern software tool
Freedom means Ability to work and think freely
I would build Software and share it openly so others can benefit and improve it
I think knowledge grows when it is shared, and open source is the basis of that belief
```

Explanation:

The script is written to generate an open-source manifesto by taking input from the user.

The following questions are asked by the program:

Name one open-source tool you use every day:

In one word, what does 'freedom' mean to you?

Name one thing you would build and share freely:

The responses received from the user are written as a single paragraph. This content is saved into a text file The name of the file is given when the program executes, and the date is saved using the "date" command.

It shows the manner in which output redirection can be used to input content into the file, and comments are included to show the use of aliases to make command execution simple. The concepts used are read for user input, string concatenation, writing to a file with >, the date command, and the aliases concept demonstrated via a comment.

Conclusion:

This project has helped me to get a deeper understanding of Open-Source software Python.

It has given me a deeper understanding of Open-Source software and Linux Systems.

By coding various shell scripts, I have understood various concepts like variables, loops, if-else statements and many other coding concepts. Earlier, these were very confusing to me, but while doing this project, I got a deeper understanding of these topics. Apart from this, I also realised why Python is one of the most preferred programming languages.

My view of Open-Source software has also changed. I used to think it was not safe and that it did not lead to ideas. Now I know that is not true. I also learned about some features of Python. This helped me understand where Python is used.

This project helped me understand Linux and how to write shell scripts with Python.

This project has improved my understanding of Linux and writing shell scripts using Python as a software tool.